

Instructions. (36 points) Sections 14.6–14.7. Show relevant work for full credit.

(8^{pts}) 1. Let $f(x, y) = x^2y + \sqrt{y}$.

(a) (3 pts) Find $\nabla f(x, y)$ and evaluate it at the point $(2, 1)$.

Solution:

$$\nabla f = \left\langle 2xy, x^2 + \frac{1}{2\sqrt{y}} \right\rangle$$

At $(2, 1)$:

$$\nabla f(2, 1) = \left\langle 2(2)(1), 4 + \frac{1}{2} \right\rangle = \boxed{\left\langle 4, \frac{9}{2} \right\rangle}$$

(b) (3 pts) Find the directional derivative of f at $(2, 1)$ in the direction of the vector $\mathbf{v} = \langle 3, -4 \rangle$.

Solution:

Unit vector: $\hat{u} = \frac{1}{5}\langle 3, -4 \rangle$.

$$D_{\hat{u}}f(2, 1) = \nabla f(2, 1) \cdot \hat{u} = \left\langle 4, \frac{9}{2} \right\rangle \cdot \frac{1}{5}\langle 3, -4 \rangle = \frac{1}{5}(12 - 18) = \boxed{-\frac{6}{5}}$$

(c) (2 pts) What is the maximum rate of change of f at $(2, 1)$?

Solution:

$$|\nabla f(2, 1)| = \left| \left\langle 4, \frac{9}{2} \right\rangle \right| = \sqrt{16 + \frac{81}{4}} = \sqrt{\frac{145}{4}} = \boxed{\frac{\sqrt{145}}{2}}$$

(6^{pts}) **2.** Find the equation of the tangent plane to the surface

$$x^2 + 2y^2 + 3z^2 = 12$$

at the point $(2, 1, 1)$.

Hint: Write $F(x, y, z) = x^2 + 2y^2 + 3z^2$ and use ∇F .

Solution:

Let $F(x, y, z) = x^2 + 2y^2 + 3z^2$. Then:

$$\nabla F = \langle 2x, 4y, 6z \rangle$$

At $(2, 1, 1)$:

$$\nabla F(2, 1, 1) = \langle 4, 4, 6 \rangle$$

Tangent plane:

$$4(x - 2) + 4(y - 1) + 6(z - 1) = 0$$

$$\boxed{4x + 4y + 6z = 18} \quad \text{or equivalently } 2x + 2y + 3z = 9$$

(8pts) **3.** Find all critical points of $f(x, y) = x^3 - 6xy + 8y^3$ and classify each as a local maximum, local minimum, or saddle point.

(a) (4 pts) Find the critical points.

Solution:

$$f_x = 3x^2 - 6y = 0 \Rightarrow y = \frac{x^2}{2}$$

$$f_y = -6x + 24y^2 = 0 \Rightarrow x = 4y^2$$

Substituting $x = 4y^2$ into $y = x^2/2$:

$$y = \frac{(4y^2)^2}{2} = 8y^4 \Rightarrow y(8y^3 - 1) = 0$$

So $y = 0$ (giving $x = 0$) or $y = \frac{1}{8}$ (giving $x = 1$).

Critical points: $(0, 0)$ and $(1, \frac{1}{8})$.

(b) (4 pts) Classify each critical point using the Second Derivative Test.

Solution:

$$f_{xx} = 6x, \quad f_{yy} = 48y, \quad f_{xy} = -6.$$

$$D = f_{xx}f_{yy} - (f_{xy})^2 = 288xy - 36.$$

At $(0, 0)$: $D = 0 - 36 = -36 < 0 \Rightarrow$ saddle point.

At $(1, \frac{1}{8})$: $D = 288 \cdot \frac{1}{8} - 36 = 144 - 36 = 108 > 0$ and $f_{xx} = 6 > 0 \Rightarrow$ local minimum.

(8^{pts}) 4. Find the absolute maximum and absolute minimum values of $f(x, y) = x + y - xy$ on the triangular region D with vertices $(0, 0)$, $(6, 0)$, and $(0, 6)$.

(a) (3 pts) Find any critical points of f in the interior of D .

Solution:

$$f_x = 1 - y = 0 \Rightarrow y = 1. \quad f_y = 1 - x = 0 \Rightarrow x = 1.$$

Critical point: $(1, 1)$. Check: $1 + 1 < 6$ so it is inside D . $f(1, 1) = 1 + 1 - 1 = \boxed{1}$.

(b) (5 pts) Find the extreme values of f on each edge of the boundary and determine the absolute maximum and minimum on D .

Solution:

Edge 1: $y = 0$, $0 \leq x \leq 6$. $f(x, 0) = x$. Min = 0 at $(0, 0)$; max = 6 at $(6, 0)$.

Edge 2: $x = 0$, $0 \leq y \leq 6$. $f(0, y) = y$. Min = 0 at $(0, 0)$; max = 6 at $(0, 6)$.

Edge 3: $x + y = 6$, i.e., $y = 6 - x$, $0 \leq x \leq 6$.

$$f(x, 6 - x) = x + (6 - x) - x(6 - x) = 6 - 6x + x^2$$

$$f'(x) = -6 + 2x = 0 \Rightarrow x = 3, \quad y = 3$$

$$f(3, 3) = 3 + 3 - 9 = -3.$$

Comparing all values:

Point	f value
$(0, 0)$	0
$(6, 0)$	6
$(0, 6)$	6
$(1, 1)$ interior	1
$(3, 3)$ boundary	-3

$\boxed{\text{Absolute max} = 6}$ at $(6, 0)$ and $(0, 6)$. $\boxed{\text{Absolute min} = -3}$ at $(3, 3)$.

- (6^{pts}) 5. A cardboard box without a lid is to have a volume of 32,000 cm³. Find the dimensions that minimize the amount of cardboard used.

Hint: Let the base be $x \times y$ and the height be z . Minimize the surface area $S = xy + 2xz + 2yz$ subject to $xyz = 32,000$.

Solution:

Constraint: $z = \frac{32000}{xy}$. Substitute into S :

$$S(x, y) = xy + 2x \cdot \frac{32000}{xy} + 2y \cdot \frac{32000}{xy} = xy + \frac{64000}{y} + \frac{64000}{x}$$

$$S_x = y - \frac{64000}{x^2} = 0 \quad \Rightarrow \quad x^2 y = 64000$$

$$S_y = x - \frac{64000}{y^2} = 0 \quad \Rightarrow \quad xy^2 = 64000$$

From these: $x^2 y = xy^2$, so $x = y$.

Then $x^2 \cdot x = 64000 \Rightarrow x^3 = 64000 \Rightarrow x = 40$.

So $x = y = 40$ and $z = \frac{32000}{1600} = 20$.

Dimensions: $40 \times 40 \times 20$ cm